Name: Nguyễn Dương Phúc
Student ID: 24521386
Class: CS4323.Q12.CTTT

OS
LAB 1

CHECKLIST

1. Homework

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Mark: 10/10 (self-scrores)

**Notice: Export PDF file, name format:*
<MSSV>_LAB1.pdf

2. Lab 1

1. Execute, google, and explain usage of all commands in this section

Answer:

1. Syntax: **ls** [options] [path]

```
yangpuck@NguyenDuongPhuc:~$ ls
Desktop    Downloads          Music     Public     Videos
Documents  examples.desktop   Pictures  Templates
```

Function: Lists files and directories in the current directory or a specified path.

2. Syntax: **cd** [path]

```
yangpuck@NguyenDuongPhuc:~$ cd Downloads
yangpuck@NguyenDuongPhuc:~/Downloads$ cd ..
yangpuck@NguyenDuongPhuc:~$ cd /bin
yangpuck@NguyenDuongPhuc:/bin$ cd
yangpuck@NguyenDuongPhuc:~$
```

Function: Changes the current working directory.

- **cd Downloads**: Go into the Downloads folder (move into a subdirectory).
- **cd ..** : Go up one directory level (to the parent folder).
- **cd /bin** : Go directly to the /bin directory (absolute path from root).
- **cd** : Go to your home directory (~).

3. Syntax: **pwd**

```
yangpuck@NguyenDuongPhuc:~/Downloads/I/am/Standing/Here$ pwd
/home/yangpuck/Downloads/I/am/Standing/Here
```

Function: Displays the full path of the current directory.

4. Syntax: **touch** filename

```
yangpuck@NguyenDuongPhuc:~$ ls
Desktop    Downloads          Music     Public     Videos
Documents  examples.desktop   Pictures  Templates
yangpuck@NguyenDuongPhuc:~$ touch New.txt
yangpuck@NguyenDuongPhuc:~$ ls
Desktop    Downloads          Music     Pictures   Templates
Documents  examples.desktop   New.txt   Public     Videos
```

Function: Creates an empty file or updates the timestamp if the file already exists.

5. Syntax: **nano** filename

```
yangpuck@NguyenDuongPhuc:~$ nano New.txt
  GNU nano 2.9.3                    New.txt                    Modified

you can write anything in here

^G Get Help    ^O Write Out   ^W Where Is    ^K Cut Text    ^J Justify
^X Exit        ^R Read File   ^\ Replace     ^U Uncut Text  ^T To Spell
```

Function: Opens a terminal-based text editor to edit or create a text file.

6. Syntax: **mkdir** [options] directory_name

```
yangpuck@NguyenDuongPhuc:~/Downloads$ mkdir NewFolder
yangpuck@NguyenDuongPhuc:~/Downloads$ ls
NewFolder
```

Function: Creates new directories.

7. Syntax: **rm** [options] filename_or_directory

```
yangpuck@NguyenDuongPhuc:~/Downloads$ ls
NewFolder
yangpuck@NguyenDuongPhuc:~/Downloads$ rm -r NewFolder
yangpuck@NguyenDuongPhuc:~/Downloads$ ls
yangpuck@NguyenDuongPhuc:~/Downloads$
```

```
Desktop    Downloads          Music     Pictures  Templates
Documents  examples.desktop   New.txt   Public    Videos
yangpuck@NguyenDuongPhuc:~$ rm New.txt
yangpuck@NguyenDuongPhuc:~$ ls
Desktop    Downloads          Music     Public    Videos
Documents  examples.desktop   Pictures  Templates
```

Function: Deletes files or directories.

- r → recursive (for directories)
- f → force (no confirmation)

8. Syntax: **cp** [options] source destination

```
original
yangpuck@NguyenDuongPhuc:~/Downloads$ cp original copy
yangpuck@NguyenDuongPhuc:~/Downloads$ ls
copy  original
```

Function: Copies files or directories.

9. Syntax: **mv** source destination

```
yangpuck@NguyenDuongPhuc:~$ mv MoveFolder Downloads
yangpuck@NguyenDuongPhuc:~$ cd Downloads
yangpuck@NguyenDuongPhuc:~/Downloads$ ls
MoveFolder
```

Function: Moves files/directories or renames them.

10. Syntax: **echo** [text]

```
yangpuck@NguyenDuongPhuc:~/Downloads$ echo "Hello World!" > Myfile.txt
```

Function: Prints text or variables to the terminal.

11. Syntax: **cat** [file]

```
yangpuck@NguyenDuongPhuc:~/Downloads$ cat Myfile.txt
Hello World!
```

Function: Displays the content of files in the terminal or combines files.

2. Create a folder tree like image below:

```
Myweb
  |-images
  |     |-- icon
  |     |-- background
  |     |-- animation
  |-databases
  |-scripts
  |-java
```

Answer:

Step 1: Create Myweb and enter it

```
yangpuck@NguyenDuongPhuc:~$ mkdir Myweb
yangpuck@NguyenDuongPhuc:~$ cd Myweb/
```

Step 2: Create main subfolders and enter images

```
yangpuck@NguyenDuongPhuc:~/Myweb$ mkdir images databases scripts java
yangpuck@NguyenDuongPhuc:~/Myweb$ cd images/
```

Step 3: Create subfolders inside images

```
yangpuck@NguyenDuongPhuc:~/Myweb/images$ mkdir icon background animation
```

Result:

```
yangpuck@NguyenDuongPhuc:~$ ls -R Myweb
Myweb:
databases  images  java  scripts

Myweb/databases:

Myweb/images:
animation  background  icon

Myweb/images/animation:

Myweb/images/background:

Myweb/images/icon:

Myweb/java:

Myweb/scripts:
```

3. Google how to find files with the ".html" extension in your entire system. Copy one of them to Myweb.

Answer:

Step 1: Type the following command to find all .html files:

```
yangpuck@NguyenDuongPhuc:~$ find / -name *.html
```

- Find: the command used to search for files or directories
- /: the starting directory for the search (root directory, i.e., the whole system)
- -name: option to specify that you are searching by file name
- "*.html": pattern to match file names ending with .html

Step 2: Choose any file path from the search results. For example:

```
/usr/share/doc/libxml-libxml-pe
/usr/share/doc/time/time.html
/usr/share/doc/lintian/api.html
```

Step 3: Use the cp command to copy the file into Myweb folder:

```
yangpuck@NguyenDuongPhuc:~$ cp /usr/share/doc/time/time.html ~/Myweb/
```

4. Remove the directory Myweb/scripts.

Answer:

```
yangpuck@NguyenDuongPhuc:~$ rm -rf ~/Myweb/scripts
yangpuck@NguyenDuongPhuc:~$ ls Myweb
databases  images  java  time.html
```

- **rm**: the command used to remove files or directories
- **-r**: option to remove directories recursively (including all files and subdirectories)
- **~/Myweb/scripts**: the path of the directory to be deleted

5. Set the permission for Myweb/databases as below:

- The owner has all permissions
- Other users have no permission

Answer:

```
yangpuck@NguyenDuongPhuc:~$ chmod 700 ~/Myweb/databases
```

- **Chmod**: the command used to change file or directory permissions
- **700**: permission code where:
  - 7 = read, write, and execute for the owner
  - 0 = no permissions for the group
  - 0 = no permissions for others
- **~/Myweb/databases**: the path of the directory to set permissions

```
yangpuck@NguyenDuongPhuc:~$ ls -ld ~/Myweb/databases
drwx------ 2 yangpuck yangpuck 4096 Oct 26 07:32 /home/yangpuck/Myweb/databases
```

- **ls**: list files and directories
- **-l**: use long listing format to show detailed information
- **~/Myweb**: the directory whose contents you want to check

Result:

- **drwx------**: permission string
  - **d**: directory
  - **rwx**: owner (user) has read, write, execute
  - **---**: group has no permission
  - **---**: others have no permission
- **2**: number of links
- **yangpuck**: owner name
- **yangpuck**: group name
- **4096**: size in bytes
- **Oct 26 07:32**: last modified date and time
- **/home/yangpuck/Myweb/databases**: directory name